A REPORT

ON

"JAVA FULL STACK DEVELOPMENT"

Submitted by

Mr. Sudeep Kammar - 20241ISE00052

Under the guidance of,

Thummalapalli Lakshmi Vara Prasad

in partial fulfillment for the award of the

degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE AND ENGINEERING
(COMPUTER ENGINEERING)

AT

PRESIDENCY UNIVERSITY

BENGALURU

AUGUST 2026

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that the report of **CSE7000 – Internship** on **“JAVA FULL STACK”** being submitted by “**Sudeep Kammar**” bearing roll number “**20241ISE0052**” in partial fulfillment of the requirement for the award of the degree of Bachelor of Technology in **Computer Science and Engineering (Computer Science and Engineering)** is a Bonafide work carried out under our supervision.

Mr. Thummalapalli Lakshmi Vara Prasad
Designation
PSCS
Presidency University

Ms. Battula Bhavya
Designation
PSCS
Presidency University

Mr. Pankaj Suman
<Program Internship Coordinator>
PSCS
Presidency University

Dr. Bhuvaneshwari Patil
School level coordinator
PSCS
Presidency University

Dr. Gopal Shyam
HoD
PSCS
Presidency University

Dr. Duraipandian N
Dean – PSCS & PSIS
Presidency University

PRESIDENCY UNIVERSITY

PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING

DECLARATION

I hereby declare that the work, which is being presented in the report entitled "**JAVA FULL STACK**" in partial fulfillment for the award of Degree of **Bachelor of Technology** in **Computer Science and Engineering (Computer Science and Engineering),** is a record of my own investigations carried under the guidance of **Mr. Thummalapalli Lakshmi Vara Prasad,** Designation and **Ms. Battula Bhavya,** Designation, Presidency School of Computer Science and Engineering, Presidency University, Bengaluru. Designation and Dr. Murali Parameswaran, Designation,

I have not submitted the matter presented in this report anywhere for the award of any other Degree.

Sudeep Kammar,
20241ISE0052,

INTERNSHIP COMPLETION CERTIFICATE

Abeyaantrix

U72900KA2

CORPORATE61252a401df901629825600

CERTIFICATE

OF COMPLETION

This is to certify that

Mr. SUDEEP KAMMAR

USN:20241ISE0052, Department of B.Tech , ISE (AI & Robotics),

Presidency University, Bengaluru

has successfully completed **2 Months INTERNSHIP/ VOCATIONAL TRAINING** on

" Full Stack JAVA"

at Abeyaantrix Edusoft, Davanagere.

from **12th June 2026** to **7th August 2026**.

Shivaraj. V.S

Mr. Shivaraj V S
Manager
Abeyaantrix Edusoft LLP

Virupakshappa

Mr. Virupakshappa H.S.
Director
Abeyaantrix Edusoft LLP

www.abeyaantrix.org

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Sudeep Kammar,

20241ISE0052,

ABSTRACT

Java is one of the most widely used, high-level, object-oriented programming languages in the software industry today, valued for its platform independence, robustness, security, and scalability. This report presents a detailed account of the internship training undertaken on "Object Oriented Programming in Java", covering the language's fundamentals as well as its advanced object-oriented features.

The core of the internship focused on the four pillars of Object-Oriented Programming — Encapsulation, Inheritance, Polymorphism, and Abstraction — along with related concepts such as access specifiers, constructors, interfaces, and memory management. The training further extended into exception handling for robust error management, multithreading for concurrent programming, and the Java Collections Framework for efficient data manipulation.

To reinforce the concepts learned, two mini projects were developed during the internship: a "Student Management System" and a "Simple Banking System", both of which apply core OOP principles, exception handling, and collection classes in a practical, real-world context. This report documents the concepts studied, the implementation details of the mini projects, the outcomes achieved, and the key learnings gained from the internship experience.

1. INTRODUCTION TO INTERNSHIP

Internship training forms a vital bridge between academic learning and industry practice. It provides students with hands-on exposure to real-world technologies, tools, and development practices that are not always covered in depth within a classroom setting. As part of the academic curriculum, this internship was undertaken to gain practical proficiency in Java programming, with a strong emphasis on Object-Oriented Programming (OOP) principles.

The internship was carried out at **Abeyaantrix, Davanagere** , an organization engaged in providing technical training and skill development programs in areas such as programming languages, software development, and emerging technologies. The organization offers structured, industry-oriented training modules designed to help students and professionals build strong programming foundations.

2. OBJECTIVES OF INTERNSHIP

- To understand the fundamentals of the Java programming language and its ecosystem.
- To gain a strong conceptual and practical understanding of Object-Oriented Programming principles.
- To learn how Java programs are written, compiled, and executed using JDK, JRE, and JVM.
- To study control statements, arrays, strings, and other core programming constructs.
- To understand advanced topics such as exception handling, multithreading, and collections.
- To apply the concepts learned in the development of real-world mini projects.
- To develop problem-solving and logical thinking skills through hands-on coding practice.

3. PROGRESS

To consolidate the theoretical and practical knowledge gained during the internship, two mini projects were designed and developed. Both projects apply core Object-Oriented Programming concepts such as classes, objects, encapsulation, and constructors, together with exception handling, arrays/collections, and the Scanner class for user interaction.

3.1 Mini Project : Student Management System

Objective: To design a console-based application that allows an administrator to add, view, search, update, and delete student records, demonstrating the use of classes, objects, encapsulation, collections (ArrayList), and exception handling.

Key OOP Concepts Applied:

- Encapsulation — student details (name, roll number, marks) are kept private with public getters/setters.
- Class & Object — a Student class models each student record as an object.
- Collections Framework — an ArrayList<Student> is used to dynamically store multiple records.
- Exception Handling — invalid inputs (e.g., non-numeric roll numbers) are handled gracefully.

Sample Code:

```
import java.util.*;

class Student {
    private int rollNo;
    private String name;
    private double marks;

    public Student(int rollNo, String name, double marks) {
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
    }
    public int getRollNo() { return rollNo; }
    public String getName() { return name; }
    public double getMarks() { return marks; }
```

```java
    public void display() {
        System.out.println(rollNo + "\t" + name + "\t" + marks);
    }
}

public class StudentManagementSystem {
    static ArrayList<Student> students = new ArrayList<>();
    static Scanner sc = new Scanner(System.in);

    public static void addStudent() {
        try {
            System.out.print("Enter Roll No: ");
            int roll = Integer.parseInt(sc.nextLine());
            System.out.print("Enter Name: ");
            String name = sc.nextLine();
            System.out.print("Enter Marks: ");
            double marks = Double.parseDouble(sc.nextLine());
            students.add(new Student(roll, name, marks));
            System.out.println("Student added successfully!");
        } catch (NumberFormatException e) {
            System.out.println("Invalid input. Please enter valid data.");
        }
    }

    public static void viewStudents() {
        if (students.isEmpty()) {
            System.out.println("No records found.");
            return;
        }
        for (Student s : students) s.display();
    }

    public static void searchStudent(int rollNo) {
        for (Student s : students) {
            if (s.getRollNo() == rollNo) {
                s.display();
                return;
            }
        }
        System.out.println("Student not found.");
    }

    public static void main(String[] args) {
        int choice;
        do {
            System.out.println("\n1.Add  2.View  3.Search  4.Exit");
            System.out.print("Enter choice: ");
            choice = Integer.parseInt(sc.nextLine());
            switch (choice) {
                case 1 -> addStudent();
                case 2 -> viewStudents();
                case 3 -> {
                    System.out.print("Enter Roll No to search: ");
                    searchStudent(Integer.parseInt(sc.nextLine()));
                }
                case 4 -> System.out.println("Exiting...");
```

```
            default -> System.out.println("Invalid choice.");
        }
    } while (choice != 4);
  }
}
```

Sample Output:

```
1.Add  2.View  3.Search  4.Exit
Enter choice: 1
Enter Roll No: 101
Enter Name: Aditi Rao
Enter Marks: 88.5
Student added successfully!
```

4. LEARNING OUTCOMES

Through the course of this internship, the following key learning outcomes were achieved:

- Gained a strong foundational understanding of the Java programming language, including its syntax, data types, control structures, and program execution flow.
- Developed a clear understanding of the working of JDK, JRE, and JVM, and how Java achieves platform independence.
- Acquired practical proficiency in applying the four pillars of Object-Oriented Programming — Encapsulation, Inheritance, Polymorphism, and Abstraction.
- Learned to design robust applications using exception handling to manage runtime errors gracefully.
- Understood the fundamentals of multithreading and concurrent programming in Java.
- Learned to use the Java Collections Framework for efficient storage and manipulation of grouped data.
- Enhanced problem-solving and logical reasoning skills by translating real-world requirements into working Java programs.
- Gained hands-on experience in building complete console-based applications from requirement analysis to testing.

5. CONCLUSION

This internship on "Object Oriented Programming in Java" provided a comprehensive and structured learning experience, covering both the fundamentals of the Java language and the core principles of Object-Oriented Programming. Starting from the basics of Java syntax and the working of the JVM, the training progressed systematically through control statements, arrays, strings, and eventually into the deeper object-oriented concepts of encapsulation, inheritance, polymorphism, and abstraction.

The additional exploration of exception handling, multithreading, and the Java Collections Framework broadened the understanding of how robust, efficient, and concurrent applications are built in Java. The development of the two mini projects — the Student Management System and the Simple Banking System — served as an effective practical exercise, allowing the application of theoretical concepts to solve real-world problems.

Overall, this internship has significantly strengthened programming fundamentals and object-oriented design skills, and has laid a solid foundation for undertaking more advanced software development and engineering coursework in the future.

6. REFERENCES

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